The empirical recurrence for OEIS A211114, proved by a one-dimensional reduction

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A211114 counts symmetric $(n+1) \times (n+1)$ matrices over $\{-2, \ldots, 2\}$ in which every 2×2 subblock sums to zero and has one or three distinct entries. Both sides of the matrix grow with n, so there is no slice of bounded width to walk along. There does not need to be: the sum-zero condition is linear and solves completely, and its solutions are indexed by a SEQUENCE — the diagonal of the matrix. The count becomes a walk on $S = 12$ states, and the empirical recurrence of order 5 contributed by R. H. Hardin follows.

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1 The conjecture

OEIS A211114 is "Number of (n+1) X (n+1) -2..2 symmetric matrices with every 2 X 2 subblock having sum zero and one or three distinct values.". It has offset 1 and begins

$$9, 19, 39, 81, 167, 341, 695, 1405, \ \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 2*a(n-1) + 3*a(n-2) - 6*a(n-3) - 2*a(n-4) + 4*a(n-5).

As of the "Last modified" line on the live entry (Jul 15 2018 12:03:54, revision 10) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The sum-zero condition solved

Let x be an $N \times N$ symmetric matrix with entries in $\{-2, \ldots, 2\}$ such that

$$x(i, j) + x(i, j{+}1) + x(i{+}1, j) + x(i{+}1, j{+}1) = 0$$

for every $0 \leq i, j \leq N - 2$.

Lemma 1. *Such matrices are exactly the matrices*

$$x(i, j) = (-1)^{i+j}\, \frac{u_i + u_j}{2}, \qquad u_0, \ldots, u_{N-1} \in \{-2, \ldots, 2\} \text{ all of the same parity,}$$

and the sequence is recovered from the matrix by $u_i = x(i, i)$, so the correspondence is a bijection.

Proof. Put $y(i,j) = (-1)^{i+j} x(i,j)$. The four entries of a 2×2 subblock alternate in sign under this substitution, so

$$x(i,j)+x(i,j+1)+x(i+1,j)+x(i+1,j+1) = (-1)^{i+j}\big[y(i,j)-y(i,j+1)-y(i+1,j)+y(i+1,j+1)\big],$$

and the condition says exactly that the mixed second difference of y vanishes everywhere. Summing that identity over a rectangle gives $y(i,j) = y(i,0) + y(0,j) - y(0,0)$ for all i,j, which is $y(i,j) = f(i) + g(j)$ with $f(i) = y(i,0) - y(0,0)$ and $g(j) = y(0,j)$; conversely every such y has vanishing mixed difference. Symmetry of x gives $y(i,j) = y(j,i)$, hence $f(i) - g(i) = f(j) - g(j)$ for all i,j: the difference is a constant c. Writing $u_i = 2\big(f(i) - c/2\big)$, which is an integer because $y(i,i) = x(i,i)$ is, we get $y(i,j) = (u_i + u_j)/2$ as claimed. Taking $j = i$ gives $u_i = y(i,i) = x(i,i)$.

For the two side conditions: $u_i + u_j$ must be even for all i,j, which says all the u_i have the same parity; and $|x(i,j)| = |u_i + u_j|/2 \le 2$ for all i,j is equivalent to $|u_i| \le 2$ for each i, the case $j = i$ giving one direction and the triangle inequality the other. □

So the matrices of size N correspond one for one to the sequences $u_0, \dots, u_{N-1}$ in $\{-2, \dots, 2\}$ with all entries of one parity. What remains is to say, in terms of u, that every 2×2 subblock has one or three distinct entries.

Lemma 2. *Write $\pi_i = (u_i, u_{i+1})$ for the i-th consecutive pair. The number of distinct entries of the subblock at (i,j) depends only on π_i and π_j: with $\pi_i = (\alpha, \beta)$ and $\pi_j = (\gamma, \delta)$ it is*

$$\left|\left\{ \tfrac{\alpha+\gamma}{2},\ -\tfrac{\alpha+\delta}{2},\ -\tfrac{\beta+\gamma}{2},\ \tfrac{\beta+\delta}{2} \right\}\right|.$$

Proof. By Lemma 1 the four entries are $(-1)^{i+j}$ times those four numbers, in order. Multiplying all four by ± 1 does not change how many are distinct. □

Call two consecutive pairs COMPATIBLE when that number lies in $\{1, 3\}$. The condition on the matrix is that π_i and π_j are compatible for every i and j — including $i = j$, which asks each pair to be compatible with itself. In other words:

Corollary 3. *The matrices counted are exactly the sequences u whose set of consecutive pairs is a clique in the compatibility graph on the ≤ 25 pairs.*

3 A walk on the sequence

The clique condition is not local: a pair must be compatible with every pair anywhere else in the sequence. Carrying the set of pairs already used would be an exponential state. Carrying what it FORBIDS is not.

Lemma 4. *Let A be the set of pairs compatible with every pair used so far. Then a sequence may be continued exactly by pairs lying in A, and appending a pair π replaces A by $A \cap N(\pi)$, where $N(\pi)$ is the set of pairs compatible with π. Hence the pair (last value of u, A) carries everything the future needs.*

Proof. The condition is a conjunction of pairwise compatibilities, so the set of pairs still allowed after using a set T is $\bigcap_{\pi \in T} N(\pi)$, which is a function of T and is updated by intersection as T grows. The next pair is (last value, w), so the last value is the only other thing a step needs. □

The sets A that arise are intersections of neighbourhoods, of which there are far fewer than subsets; enumerated from the start $A_0 = \{\pi : \pi \in N(\pi)\}$, the reachable states number $S = 12$. The two parity classes never mix, so the walk starts by choosing a parity and a first value, and every state may end a sequence:

$$a(n) \;=\; \iota^{\top} M^{j} \tau, \qquad \tau = \mathbf{1}, \qquad S = 12.$$

The walk index j is the size of the matrix, and it differs from the entry's own n by a constant read off by matching the published terms rather than assumed. In particular a is C-finite — for a family in which both dimensions grow with n.

4 The proof

Lemma 5. *Let $q(t) = t^5 - \sum_i c_i t^{5-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M)\tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^\top M^j q(M)\tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+5} - \sum_i c_i M^{j+5-i}$. For the bound, M^S is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. □

Theorem 6.

$$a(n) \;=\; 2\,a(n-1) + 3\,a(n-2) - 6\,a(n-3) - 2\,a(n-4) + 4\,a(n-5)$$

for every $n > 4$.

Proof. The vector $w = q(M)\tau$ was formed by 5 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \ldots,$ again exactly; those integers vanish from the index corresponding to $n = 4 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. □

5 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 30 terms the entry publishes, exactly, in integer arithmetic.

Second, the count was recomputed for the smallest matrices straight from the definition: symmetric matrices were written out entry by entry over $\{-2, \ldots, 2\}$ and each 2×2 subblock tested for summing to zero and for having one or three distinct entries. The sequence u, the sign pattern, the clique and the state play no part in that computation, and the published terms came back. Lemma 1 was separately checked the same way: for small sizes and small ranges the number of symmetric sum-zero matrices found by direct enumeration equals the number of admissible sequences.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 5 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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